Dai (Ethan) Shi

Machine Learning Research Scientist

Generative Models | Reinforcement Learning | Stochastic Analysis | Operator Learning | Graph Neural Networks | Multimodal LLMs | Mechanistic Interpretability

University of Cambridge, United Kingdom | ds2213@cam.ac.uk | eethanshi.github.io

OpenReview | LinkedIn

RESEARCH SUMMARY

University of Cambridge postdoctoral researcher developing **theory-grounded learning algorithms** for structured and scientific data. First author on **multiple ICML/ICLR/NeurIPS papers**. Lead projects from mathematical formulation and PyTorch implementation through multi-domain evaluation and publication, with research spanning graph learning, multimodal LLMs, generative models, neural operators, and stochastic and geometric machine learning. I was also a candidate in the Chinese reserve team for the **International Mathematical Olympiad (IMO)**.

RESEARCH EXPERIENCE AND SELECTED PROJECTS

Postdoctoral Research Fellow **March 2026–Present**

University of Cambridge; advisor: Prof. José Miguel Hernández-Lobato *Cambridge, United Kingdom*

- **S3GNN (ICML 2026 Spotlight):** co-led a lightweight global–local graph architecture with GCN-comparable layer complexity and no eigendecomposition; the method reduced error by up to an order of magnitude with up to 50% fewer parameters across long-range graph benchmarks, knowledge-graph QA, and mesh-based fluid dynamics.
- **Expanding the Chaos (ICML 2026):** developed a Wiener-chaos neural operator that reconstructs SDE/SPDE trajectories in one forward pass; evaluated the framework across scientific solvers, diffusion sampling, graph interpolation, financial extrapolation, parameter estimation, and flood forecasting.
- **B-NRDE (ICML 2026):** developed neural rough differential equation framework for Itô and manifold-valued dynamics, preserving geometric constraints across volatility, $\mathrm{SO}(3)$, and covariance forecasting tasks.
- **SPDEbench (NeurIPS 2026):** main contributor on the first benchmark paper on both non-singular/singular SPDEs, linked at https://github.com/DeepIntoStreams/SPDE_hackathon, thoroughly train/evaluate current SPDE solvers (e.g., neural operators) via multiple metrics.
- **Learning on Stochastic Dynamics (NeurIPS 2026 Workshop PC)**: Programe Chair on the workshop: *AI for Stochastic Dynamics, from Theoretical Foundation to Scientific Applications.* https://eethanshi.github.io/stochastic-dynamics-2026/.

Research Fellow **2023–2025**

University of Sydney; advisor: Prof. Junbin Gao *Sydney, Australia*

- **ATOM (ICLR 2026):** We designed a pretrained transformer-based neural operator for multi-task molecular dynamics forecasting. We also curated TG80, a large, diverse, and numerically stable MD dataset with over 2.5 million femtoseconds of trajectories across 80 compounds.
- **GNN-Diff (ICLR 2025):** graph-conditioned latent diffusion framework that generates strong GNN parameters from coarse-search checkpoints; validated it through 166 experiments spanning 10 architectures, 20 public datasets, and four graph tasks.
- **DMD-GNN (ICLR 2025):** connected graph representation learning with Koopman theory and dynamic mode decomposition; evaluated low-rank models across directed, large-scale, long-range, spatiotemporal, and link-prediction tasks.
- **ACT (NeurIPS 2025):** developed an annotator–judge–human-review pipeline for language, vision, and multimodal data; the resulting system kept downstream performance within 2% of fully human labels on most benchmarks while reducing human annotation cost by up to 90%.

TECHNICAL SKILLS

Programming and ML: Python, PyTorch, NumPy, SciPy, C++, MATLAB; model training, experimental and benchmark design, evaluation pipelines, numerical optimization, and scientific computing.
Methods: deep learning, graph neural networks, generative and diffusion models, transformers and multimodal LLM evaluation, parameter-efficient fine-tuning, neural operators, neural ODE/SDE/RDE models, and time-series forecasting.
Mathematics: stochastic analysis, spectral methods, optimal transport, Riemannian optimization, manifold learning, and numerical differential equations.

SELECTED KEY PUBLICATIONS

1. Kai Zhao*, **D.Shi***, et al. *RelGS: Relation-Aware Gaussian Splatting for Open-Vocabulary 3D Scene Understanding*. **NeurIPS 2026**.
2. Yuantu Zhu, **D.Shi**, et al. *SPDEBench: An Extensive Benchmark for Learning Stochastic PDEs*. **NeurIPS 2026**.
3. L.Zhao, **D.Shi**, et al. *LOFT: Low-Rank Orthogonal Fine-Tuning via Task-Aware Support Selection*. **NeurIPS 2026**.
4. **D. Shi**[*], L. Thompson[*], L. Luo[*], et al. *S3GNN: Efficient Global Mixing and Local Message Passing for Long-Range Graph Learning*. **ICML 2026, Spotlight (top 2%)**.
5. **D. Shi**[*], L. Lin[*], A. Han[*], L. Thompson[*], et al. *Expanding the Chaos: Neural Operator for Stochastic (Partial) Differential Equations*. **ICML 2026**.
6. L. Thompson[*], **D. Shi**[*], L. Lin, et al. *Learning Manifold and Itô Dynamics with Branched Neural Rough Differential Equations*. **ICML 2026**.

7. L. Thompson, D. Guan, S. Matthews, **D. Shi**, et al. *ATOM: A Pretrained Neural Operator for Multitask Molecular Dynamics*. **ICLR 2026**.
8. L. Lin*, **D. Shi***, A. Han, et al. *ACT as Human: Multimodal Large Language Model Data Annotation with Critical Thinking*. **NeurIPS 2025**.
9. **D. Shi***, L. Lin*, A. Han, et al. *When Graph Neural Networks Meet Dynamic Mode Decomposition*. **ICLR 2025**.
10. L. Lin*, **D. Shi***, A. Han, et al. *Diffusing to the Top: Boost Graph Neural Networks with Minimal Hyperparameter Tuning*. **ICLR 2025**.
11. R. Li*, **D. Shi***, Y. Xiao*, J. Gao. *UFGTime: Mining Intertwined Dependencies in Multivariate Time Series via an Efficient Pure Graph Approach*. **PVLDB 2025**.
12. **D. Shi***, Z. Shao*, J. Gao, Z. Wang, Y. Guo. *Frameless Graph Knowledge Distillation*. **IEEE TNNLS 2025**.
13. **D. Shi**, J. Gao, X. Hong, S. T. B. Choy, Z. Wang. *Coupling Matrix Manifolds Assisted Optimization for Optimal Transport Problems*. **Machine Learning 2021**.
14. **D. Shi**, A.Han, L. Lin, et al.. *Exposition on the over-squashing problem on GNNs: Current Methods, Benchmarks, and Challenges.*. **IEEE TPAMI 2026**.

RECENT PROJECTS AND SUBMITTED MANUSCRIPTS

1. **D. Shi**, X. Li, A.Han, J. M. Hernández-Lobato. *From Plato to Aristotle: Toward a General Theory of Representation Hypotheses.*. **ICLR 2027**.
2. **D. Shi**, X. Li, J. M. Hernández-Lobato. *When the Canonical Completion Is Wrong: Formalizing and Measuring the Jump in Large Language Models*. **ICLR 2027**.
3. **D. Shi**, A. Han, J. M. Hernández-Lobato. *Self-Confirming Superposition Traps in Reinforcement Learning*. **ICLR 2027**.
4. **D. Shi**, L. Thompson, A. Han, JB Gao, J. M. Hernández-Lobato. *Paracontrolled Neural Operator for Singular SPDEs.*. **ICLR 2027**.
5. **D. Shi**, X. Li, A. Han. *The Geometry of Reusing Random Sketches: Pathwise-Certified Riemannian Nyström–Newton Methods*. **AAAI 2027**.
6. **D. Shi**, X. Li, L. Thompson, J. M. Hernández-Lobato, A. Han. *How Do Stochastic Optimizers Select Within Flat Minima? An Intrinsic Riemannian Theory*. **AAAI 2027**.
7. X. Li, L. Thompson, **D. Shi**, J. Jiang, J. Gao, A. Han. *When Does Anchoring Help in Random Submanifold Descent? A Minimax Law on the Stiefel Manifold*. **AAAI 2027**.
8. X. Li, L. Zhao, Z. Sha, **D. Shi**, J. M. Hernández-Lobato, A. Han. *Exact Polar Is Not a Statistical Surrogate for Fixed-Depth Newton–Schulz: A Population-Risk Separation for the Spectral Core of Muon*. **AAAI 2027**.
9. Z.Tu, Zhe.Li, **D.Shi**, et al. *SPDEBench: An Extensive Benchmark for Learning Stochastic PDEs*. **NeurIPS 2026**.

EDUCATION

Doctor of Philosophy — **2020–2024**
University of Sydney — *Australia*

Master of Philosophy — **2016–2018**
University of Sydney — *Australia*

Master's Degree, Biostatistics and Infectious Disease Epidemiology — **2013–2014**
University of Hong Kong — *Hong Kong*

Bachelor of Engineering, Chemical Engineering — **2009–2013**
University of Oxford — *United Kingdom*

LEADERSHIP, RECOGNITION & SERVICE

- **Organizer, Program Chair**, NeurIPS 2026 Workshop on AI for Stochastic Dynamics: From Theoretical Foundations to Scientific Applications.
- **Team lead, prize-winning award**, 2025 Singular SPDE Modelling Hackathon.
- Reserved member for **International Mathematical Olympiad**.
- **Reviewer/Area Chair** for ICLR, ICML, NeurIPS, KDD, and VLDB; journal reviewer for *Artificial Intelligence*, JMLR, TMLR, and *Machine Learning*.